Alexander Joos

+1-978-799-9242 | aaj46@cornell.edu | linkedin.com/in/alexanderjoos1 | github.com/alexanderjoos

Education

Cornell University Ithaca, NY
MEng Computer Science, GPA: 3.65 *Jan. 2026 – Dec. 2026*
Cornell University Ithaca, NY
B.A. Information Science (Data Science) *Aug. 2022 – Dec. 2025*
National University of Singapore Singapore
Yale-NUS College Exchange Program (Computer Science) *Jan. 2025 – May 2025*
Graduate Coursework: Algorithms, Systems Programming in Rust, Distributed ML Systems (LLM Training & Inference), Software Engineering, Machine Learning, Deep Learning, LLM Applications (RAG & AI Agents), Market Microstructure

Skills

Languages: Python, SQL, Java, C/C++, Rust, OCaml, JavaScript, TLA+
ML & AI: PyTorch, Hugging Face, scikit-learn, LangChain, LLM APIs, MCP, RAG, Vector DBs, Multi-Agent Systems
Data: Kafka, Aeron, Airflow, pandas, SingleStore, Snowflake, BigQuery, PostgreSQL, Oracle, CockroachDB
Backend & Cloud: Flask, FastAPI, REST APIs, Docker, AWS (S3), GCP, Linux, Git, CI/CD, Claude Code, Cursor

Experience

Fidelity Investments June 2026 – Aug. 2026
Graduate Data Engineering Intern, Database Modernization *Merrimack, NH*
- Built Python migration agents moving COBOL-era APIs & legacy data into an event-sourced ledger via a four-pass design.
- Developed validation agents & tool-call workflows checking migrated output with TLA+/TLC specs & Python checks.
- Wrote high-throughput Java Kafka producer over Aeron with async fault-tolerant delivery for a sub-3ms brokerage platform.
- Built Aeron/Kafka pipelines streaming a deterministic brokerage platform's state machine output into SingleStore.

Fidelity Investments June 2025 – Aug. 2025
Data Engineering Intern, Cloud and Platform Engineering *Merrimack, NH*
- Owned & launched a Flask/AJAX web app on an Ubuntu VM to automate internal database model comparison.
- Designed Flask REST API endpoints & Python/SQL backend storage/retrieval behind a JavaScript/HTML/CSS frontend.
- Deployed the application through Apache, SSL certificate setup, firewall whitelisting, & DEV/PROD rollout.
- Supported brokerage platform model analysis across Oracle, PostgreSQL, CockroachDB, & Snowflake databases.

American Family Insurance: Homesite June 2023 – Aug. 2024
Partner Distribution Intern → Data Analytics Intern *Boston, MA*
- Built BigQuery SQL pipelines from raw source tables for a Fortune 50 retailer's Partner Management reporting layer.
- Built 8 Looker dashboards for Partner & Executive teams on top of the BigQuery reporting layer.
- Automated SQL query generation with Excel/Sheets scripts, replacing manual query writing for recurring reports.

Projects

Reuse-Aware Rollout Scheduling for Agentic RL – *Cornell ML Systems Research* Oct. 2026 – Present
- Instrumenting veRL async rollouts on A100s to measure tool-cache misses caused by concurrent duplicate tool calls.
- Building a trace-driven rollout simulator to test reuse-aware schedules: scout rollouts, prompt batching, hit priority.

BME Collaborator Finder – *Cornell AI Innovation Lab* Sep. 2026 – Present
- Building multi-agent collaboration system where researcher-persona agents converse to surface Cornell research partnerships.
- Developing agentic RAG foundation: PDF & PubMed ingestion into S3, vector search, & cited faculty matching.

Apache Airflow Anomaly Detection – *Agile Open Source Systems & Backend Feature Development* May 2026
- Built Airflow-native task-runtime anomaly detection with DAG-file config, custom detectors, & 4 built-in algorithms.
- Implemented backend/task SDK, structured results, task-level observability, CI tests, & example DAGs/docs.

LOB-Driven Optimal Execution Strategy – *Quantitative Trading & Market Microstructure* May 2026
- Beat TWAP on INTC/MSFT/AMZN/GOOG & held-out AAPL using order-book imbalance, spread, & timing signals.
- Generalized across names by fitting parameters per spread regime via smoothed grid search on tick-level LOB data.